

PRINCE PATEL

Software Engineering Student

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GitHub — LinkedIn — Portfolio — LeetCode

Skills

Programming: C++, C, JavaScript, Python, Java
Core CS: DSA, DBMS, OOP, Computer Networks, Operating Systems
Backend: Node.js, Express.js, JWT Authentication, REST APIs, Socket.io
Frontend: React.js, HTML, CSS
Databases: PostgreSQL, MySQL, MongoDB, pgvector
Tools: Git, GitHub, Linux, Postman, VS Code
AI/ML: XGBoost, Vector Search (pgvector), Retrieval-Augmented Generation (RAG), Feature Engineering, Machine Learning Fundamentals, LLMs

Education

Nirma University <i>B.Tech in Computer Science and Engineering</i>	2024 – 2028 <i>CGPA: 8.96</i>
Gayatri Vidhyalaya (GHSEB) <i>Class XII</i>	2024 <i>Score: 90.90%</i>

Projects

Cortex – Distributed Observability and Root Cause Platform — GitHub	Jul 2026 – Present
– Built a distributed observability platform using React, Node.js, PostgreSQL, and Socket.io to monitor latency and failures across 10 microservices, streaming telemetry at 3-second intervals. – Implemented request correlation, tracing, and dependency analysis to identify root causes across 11 simulated failure scenarios. – Developed an ML + RAG remediation pipeline by training 3 XGBoost models on 200K+ simulated telemetry records and integrating pgvector-based retrieval to generate context-aware recovery guidance.	
MiniDB – Lightweight SQL Engine with Time-Travel Queries — GitHub	Jun 2026
– Built a lightweight SQL engine from scratch in C++ supporting 10 SQL-like commands, including CRUD operations, WHERE filtering, indexing, pagination, HISTORY, and ROLLBACK. – Designed an incremental versioning mechanism enabling efficient retrieval and restoration of previous database states. – Implemented query parsing and execution using custom data structures and file-based storage.	

Hackathons

Finalist – ODOO Hackathon 2026	Jul 2026 – Aug 2026
Finalist – ODOO × KSV Hackathon 2026	Jul 2026
6th Place – GDG Hackathon, Nirma University	Jan 2026

Achievements

Reliance Foundation Scholar
Solved 290+ LeetCode problems (Peak rating: 1530)
Completed the NeetCode 150 roadmap